

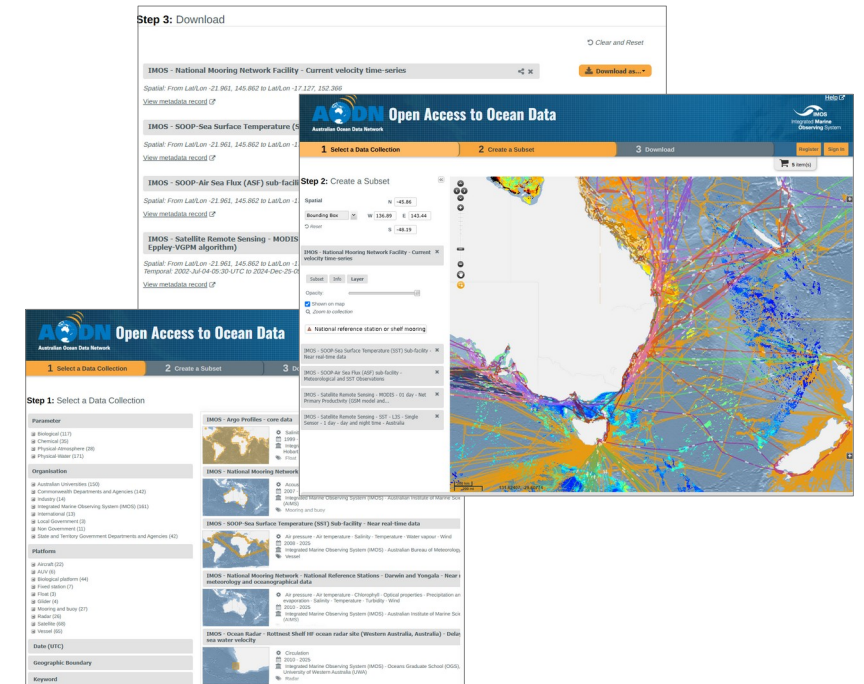
Accessing data using the AODN Portal

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Australian Ocean Data Network (AODN), Integrated Marine Observing System (IMOS)



IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

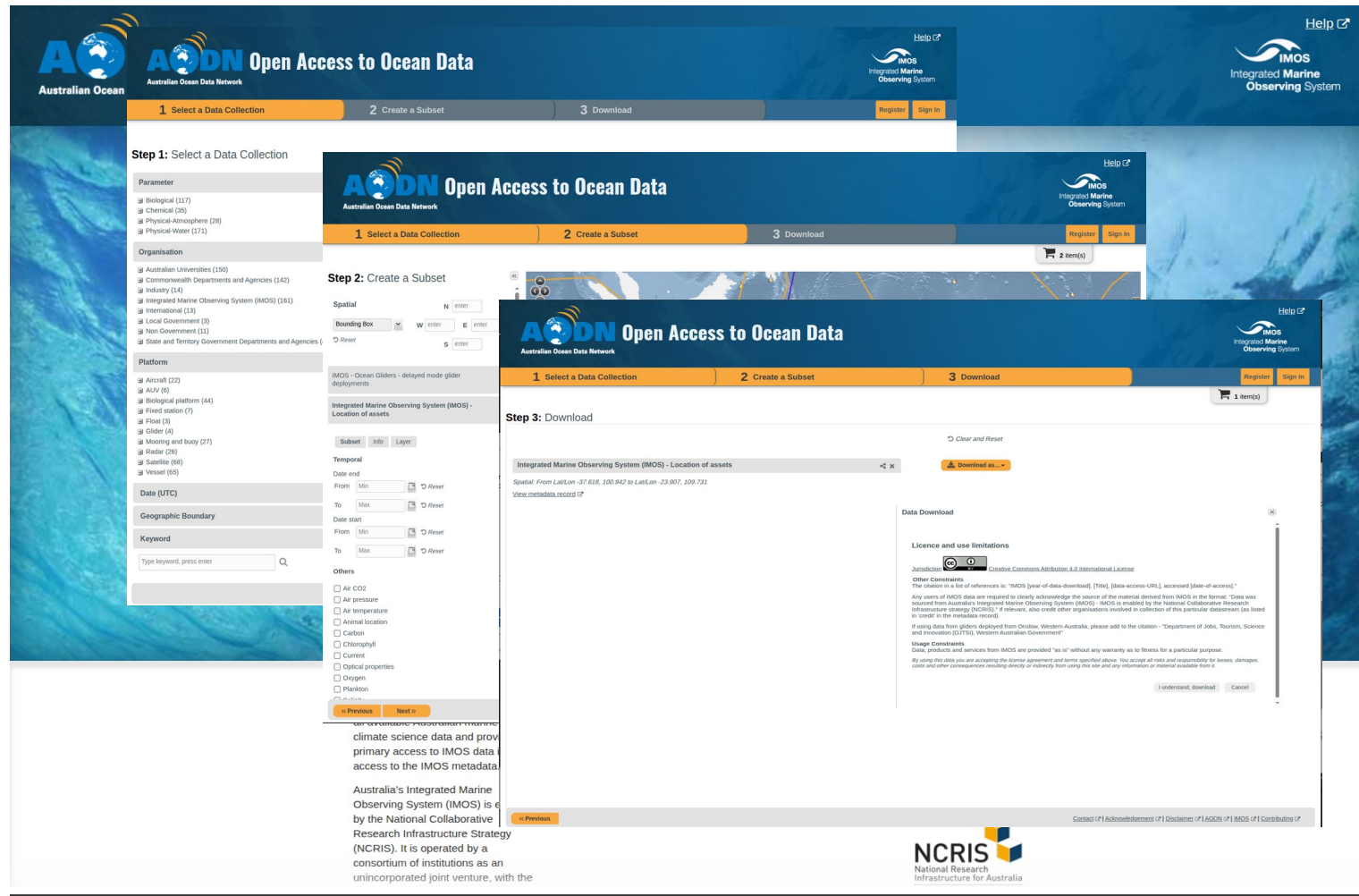
AODN Portal – A portal for the discovery of IMOS data

Outline

- Overview of the existing AODN portal
- Scenario for data access and download
- Using THREDDS server
- Additional resources

AODN Portal – Open access to ocean data

<https://portal.aodn.org.au/>



- > Latest version of the AODN portal released in 2016
- discovery and access of Australian marine data from IMOS and partner organisations.
- > Currently ~300 discoverable datasets
- > Portal designed as a simple 3 step data access source - discover, subset , download

AODN Portal – Data discovery, subset, download

– Animal tracking





AODN Open Access to Ocean Data
Australian Ocean Data Network

Help



Integrated Marine Observing System

1 Select a Data Collection

2 Create a Subset

3 Download

Guest

Register

Sign In

1 item(s)

Step 3: Download

IMOS - Animal Tracking Facility - Satellite Relay Tagging Program - Near real-time data with quality-controlled locations

Spatial: From Lat/Lon -48.92, 145.412 to Lat/Lon -45.405, 152.883

View metadata record

View profile plots from Oceancurrent SealCTDs page

Download as...

CTD Profile

CSV

Python

Argos Locations

CSV

Python

Dive Profiles

CSV

Python

Haulout

CSV

Python

SSM QC Locations

CSV

Python

Summary Statistics

CSV

Python

Subset downloaded as csv file with metadata header

# data column name	ct_standard_name	imos_vocabulary_name	unit_name	unit_short_name	ux_defn
# lat	latitude	Latitude north	Degrees	deg	
# lon	longitude	Longitude east	Degrees	deg	
# sal_corrected_val8	sea_water_salinity	Practical salinity of the water body	Practical Salinity Unit	PSU	
# sal_dbat	sea_water_pressure	Pressure (measured variable) in the water body exerted by overlying sea water and any medium above it	Decibars	dbar	
# sal_val8	sea_water_salinity	Practical salinity of the water body	Practical Salinity Unit	PSU	
# temp_dbat	sea_water_pressure	Pressure (measured variable) in the water body exerted by overlying sea water and any medium above it	Decibars	dbar	
# temp_val8	sea_water_temperature	Temperature of the water body	Degrees Celsius	degC	
#					
# device_id	device_vmpo_ref	tag_type	common_name	release_site	release
# c135-154BAT2-15	Q9901332	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-159BAT2-15	Q9901338	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-160BAT-15	Q9901329	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-163BAT-15	Q9901339	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-169BAT-15	Q9901322	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-172BAT-15	Q9901333	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-179BAT-15	Q9901335	CTD_GEN_15A	Southern elephant seal	Isle de Kerguelen	
# c135-622 BAT2-17	Q9901331	CTD_CONT_18BSD	Southern elephant seal	Isle de Kerguelen	
# c135-668BAT2-14	Q9901337	CTD_GEN_18B	Southern elephant seal	Isle de Kerguelen	
# c157-C182-20	Q9901291	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C183-20	Q9901330	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C184-20	Q9901326	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C185-20	Q9901309	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C186-20	Q9901323	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C188-20	Q9901321	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C189-20	Q9901313	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C190-20	Q9901292	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C191-20	Q9901293	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C201-20	Q9901294	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C202-20	Q9901334	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C203-20	Q9901295	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C204-20	Q9901296	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C205-20	Q9901327	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C206-20	Q9901297	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C207-20	Q9901316	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	
# c157-C208-20	Q9901308	CTD_GEN_18C	Southern elephant seal	Isle de Kerguelen	

Previous

A yellow autonomous underwater vehicle (AUV) is shown operating near the surface of the ocean. The AUV is yellow with a black conical nose and is positioned in the water. In the background, a white research vessel with a blue stripe is visible on the horizon under a clear blue sky. A bird is also seen flying in the upper left corner of the frame.

Python script

```
1 #! /usr/bin/perl -w; use strict; use warnings; use utf8;
2 #!/usr/bin/perl python
3 import codecs
4 import csv
5
6 try:
7     # Python 2
8     from urllib.request import urlopen
9 except ImportError:
10     # Python 3
11     from urllib2 import urlopen
12
13 # The URL to new collection url (non-operated url)
14 collection_url = "https://goosever.portal.sbbn.org/dgoseverer/wstypewebnew-csvofmg_on_trajectory/dataSERVICE=WSOutputformat=csv&REQUEST=feature&VERSION=0.0_&COL_FILTER={deployment}&INSTRUMENT=2&INSTRUMENT=2010c20250120?&sourceId=agent"
15
16 # Fetch data...
17 response = urlopen(collection_url)
18
19 # Iterate on data...
20 csvfile = csv.reader(codecs.iterdecode(response, 'utf-8'))
21
22 for row in csvfile:
23     print(row)
```


Extract + IMOS_-_Ocean_Gliders_-_delayed_mode_glider_deployments_source_files (1).zip

Location: /

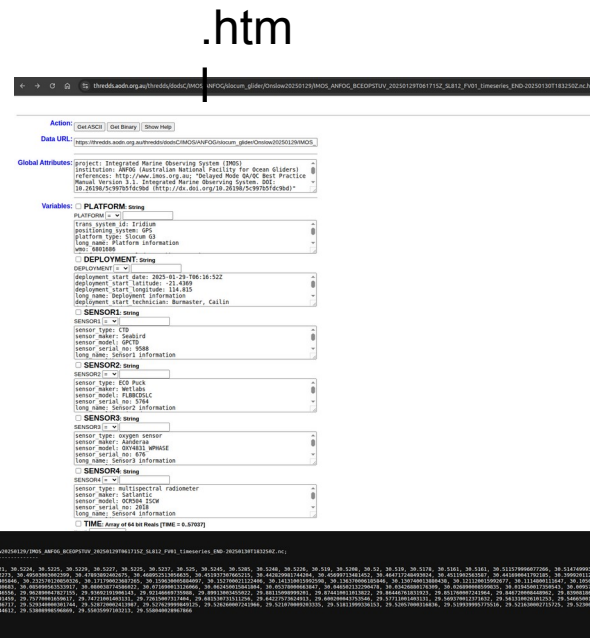
Name	Size
IMOS_ANFOG_BCEOPSTUV_20250129T061715Z_SL812_FV01_timeseries_END-20250130T183250Z.nc	5.5 MB
download_report.txt	818 bytes

```
~/Downloads/IMOS_-_Ocean_Gliders_-_delayed_mode_glider_deployments_URLs.txt
File Edit Selection Find View Goto Tools Project Preferences Help
↳ IMOS_-_Ocean_Gliders_-_delayed_mode_glider_deployments_URLs.txt x
1 https://imos-data.s3.ap-southeast-2.amazonaws.com/IMOS/ANF0G/slocum_glider/Onslow20250129/
  IMOS_ANF0G_BCEOPSTUV_20250129T061715Z_SL812_FV01_tmesieres_END_20250130T183250Z.nc
2
```

<https://thredds.aodn.org.au/thredds/catalog.html>

→ provides a data access form using standard http protocol : data subset, and interface to view metadata.

Options are available using the OpenDAP url extensions



AODN portal

Additional resources: IMOS website : <https://imos.org.au>

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I'm looking for...

Home / Data / Ocean Information Resources

Ocean Information Resources

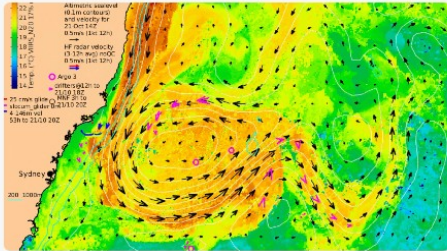
These resources provide improved accessibility to the wealth of data we collect around Australia. This includes data visualisation products, specialised data applications, toolkits for data scientists and analysis code examples.

We provide innovative methods for people to connect and use our data and understand it in different ways.

IMOS OceanCurrent

IMOS OceanCurrent is an online application to access the latest information about the state of the oceans and seas around Australia. Users can view both current and historical ocean conditions .

Learn more →



Data usage tutorials and libraries

A range of tutorials to assist users with accessing, processing and visualising data using GIS software. Python, R and MATLAB code examples are available to help people get started with using IMOS data.

Learn more →



Features links to:

- OceanCurrent
- Biological Ocean Observer
- Tutorials: GIS tutorial and user code library (Python, R, Matlab)

Data processing tools: NetCDF ninja, IMOS toolbox, Remora...

Questions?

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<https://imos.org.au/>

<https://portal.aodn.org.au/>

Vicky Isaac

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<https://oceancurrent.aodn.org.au/>

<https://shiny.csiro.au/BioOceanObserver/>

